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Future-aware blood glucose forecasting using knowledge distillation with transformer-based sequence-to-sequence models

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Accurate blood glucose level (BGL) forecasting is critical for diabetes self-management and clinical decision-making. Although deep learning models based on continuous glucose monitoring (CGM) data have achieved encouraging results, most approaches rely exclusively on historical observations and cannot explicitly account for future disturbances, such as insulin delivery and meal intake, that are unavailable at deployment. To address this limitation, we propose a future-aware learning framework for multi-step BGL prediction that leverages privileged information during training while preserving deployability at inference. A Transformer-based teacher model is trained offline using both historical CGM data and future disturbance information to learn disturbance-aware temporal representations. A student model with a similar sequence-to-sequence structure is then trained using knowledge distillation to approximate the teacher's representations based solely on historical inputs, enabling real-time forecasting without access to future data. The proposed framework is evaluated on the publicly available OhioT1DM and AZT1D datasets for prediction horizons ranging from 30 to 120 minutes and compared with several established methods. The results show consistent reductions in root mean squared error and mean absolute error, together with improved clinical reliability as assessed by Clarke error grid analysis, with over 90% of predictions falling within clinically acceptable regions. These findings demonstrate the potential of future-aware training strategies to enhance glucose forecasting performance under realistic deployment constraints.

Diabetes mellitus is a chronic metabolic condition that affects around 589 million people worldwide and is characterized by sustained hyperglycemia due to the body's inability to produce or effectively utilize insulin[1]. Type 1 diabetes mellitus (T1DM), in particular, results from autoimmune destruction of insulin-producing β -cells in the pancreas, necessitating lifelong exogenous insulin therapy. Maintaining blood glucose levels (BGLs) within a narrow euglycemic range (typically 70-180 mg/dL) is essential to prevent both acute complications (e.g., hypoglycemia, diabetic ketoacidosis) and long-term comorbidities (e.g., retinopathy, neuropathy, and cardiovascular disease)[2,3]. However, achieving optimal glycemic control remains a significant challenge, especially due to the nonlinear and delayed physiological responses to insulin, meals, stress, physical activity, and other lifestyle factors[4].

In clinical practice, individuals with T1DM rely on continuous glucose monitoring (CGM) systems and insulin pumps or multiple daily injections to regulate BGLs. However, a critical limitation lies in the inherent lags of glucose sensing and insulin kinetics. Subcutaneously infused insulin exhibits a delayed onset and prolonged action compared to endogenous insulin. Besides, CGM systems measure interstitial glucose rather than plasma glucose levels, which takes time for glucose to be transported between plasma and interstitial fluid[4–6]. But it is worth noting that the CGM readings are referred to as BGL in this work. These temporal mismatches make real-time glycemic regulation challenging and often lead to overcorrection or underestimation of insulin needs, resulting in frequent episodes of hypo- or hyperglycemia. To mitigate these risks, accurate BGL forecasting has been widely recognized as a key component in decision support systems for diabetes management.

To estimate future BGLs, a broad range of data-driven prediction models has been developed based on CGM data and related physiological signals. Traditional approaches, including autoregressive models, Kalman filters, Gaussian process regression, and support vector machines[7–12], have been widely investigated in modeling glucose dynamics. While these models offer some interpretability and robustness, they often fall short in capturing the nonlinear and multiscale dynamics in glycemic metabolism. More recently, the surge in deep

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learning research has led to the development of powerful models that can learn complex temporal dependencies directly from data[13]. The deep neural network (DNN) models for BGL prediction leveraged feedforward neural networks (FNNs) and recurrent neural networks (RNNs) such as long short-term memory (LSTM) and gated recurrent unit (GRU) architectures[14–18]. These models demonstrated improved performance over traditional methods by capturing long-range temporal dependencies and nonlinearities in glucose dynamics. Subsequently, convolutional neural networks (CNNs) and hybrid CNN-RNN architectures were introduced to learn local patterns and global trends simultaneously[19]. Further enhancements have been explored using attention mechanisms, transformer architectures, reinforcement learning, variational autoencoders, etc.[20–25].

Among these advanced techniques, the encoder-decoder framework has gained particular attention for sequence-to-sequence (seq2seq) forecasting tasks, including glucose prediction. This architecture consists of two components: an encoder that transforms the input sequence into a latent representation, and a decoder that generates the output sequence based on this representation. The use of attention mechanisms in the encoder-decoder framework has further improved performance by allowing the model to focus on the most relevant time steps in the input sequence. Transformers, originally introduced for natural language processing tasks, have recently been adapted to BGL forecasting and have shown state-of-the-art performance[21,26–28]. However, despite these advancements, a key limitation of the end-to-end deep learning models for BGL forecasting is that they rely solely on past observations and assume the future is independent of external disturbances. In real-world scenarios, unobserved future events, such as unannounced meals, unrecorded physical activity, stress, or insulin infusion, can dramatically alter glycemic trends. These future disturbances are not available at prediction time and are thus omitted from the model input, leading to systematic prediction errors, particularly during periods of rapid glucose change. Although various attempts have been made to compensate for this uncertainty using data augmentation, probabilistic modeling, or ensemble methods, the challenge of representing unseen future influences remains a fundamental bottleneck.

To address these limitations, we proposed a knowledge-distillation-based approach that improved the prediction accuracy of future BGLs by learning from future disturbances. The main contributions of this study are as follows:

(1) We develop a future-aware training framework for multi-step blood glucose level prediction that leverages privileged future disturbance information during training while maintaining deployability using historical data only at inference time.
(2) We design a Transformer-based teacher student architecture in which a teacher model learns disturbance-aware temporal representations from complete sequences containing both past and future glucose-related signals, and a student model approximates these representations using historical observations alone through knowledge distillation.
(3) We introduce a shared decoder strategy that enables consistent decoding of teacher and student latent representations into future blood glucose trajectories, facilitating effective comparison and transfer of predictive information.
(4) We conduct a comprehensive evaluation on a publicly available clinical continuous glucose monitoring dataset across multiple prediction horizons, demonstrating consistent performance improvements over baseline deep learning approaches and clinically acceptable prediction accuracy as assessed by Clarke error grid analysis.

In summary, this study presents a future-aware learning framework for multi-step blood glucose prediction that addresses the mismatch between training assumptions and deployment constraints in real-world diabetes management. By exploiting future disturbance information during offline training through a knowledge-distillation-based teacher student architecture, the proposed approach enables a deployable student model to capture future-relevant temporal representations using historical data only. Comprehensive evaluations on clinical CGM data demonstrate that the framework provides consistent improvements over baseline deep learning models across multiple prediction horizons while maintaining clinical reliability. These findings highlight both the potential and the practical limits of transferring future-aware information to history-only predictors, offering insights for the development of robust and deployable glucose forecasting systems.

Related works

The recent rapid development of wearable sensing and deep learning techniques provides novel opportunities for modeling the glucose metabolism of individuals with T1DM accurately, which is critical for diabetic management. The conventional DNN structures, including FNN, RNN, and CNN, have been widely investigated in modeling glucose metabolism and forecasting future BGLs.

Rabby et al.[29] combined the stacked LSTM model with Kalman filter, where the Kalman filter was applied to CGM readings for improving the robustness of the DNN model to sensor noise and errors. Wang et al.[30] proposed to reduce the non-stationarity of glycemic dynamics by decomposing CGM data with variational modal decomposition method and forecasting future BGLs with an LSTM model. Shuvo et al.[31] incorporated the stacked LSTM model with the multi-task learning framework to prevent overfitting. Considering that the personalized data is limited, Shen et al.[32] investigated the amount of data required for a stacked LSTM model and proposed an incrementally retrained LSTM model for BGL forecasting, where the model was finetuned as the distribution of new data shifted. Li et al.[33] and Bhargav et al.[34] investigated the performance of temporal convolutional network (TCN), which is designed for capturing temporal features in time series with a lighter computational burden, in forecasting future BGLs. Talebi et al.[35] extended the TCN-based BGL prediction model by integrating an attention mechanism, which improved the model's ability in capturing both local and global temporal correlations in glycemic metabolism. Li et al.[19] and Jaloli et al.[36] proposed to forecast future BGLs with

a hybrid CNN and LSTM structure where the CNN layers were utilized to capture local patterns in the glycemic metabolism and LSTM was adopted to model the temporal dependencies. Sun et al.[37] and Xie et al.[38] investigated the efficiency of LSTM model, Bidirectional LSTM (BiLSTM), and TCN in short-term BGL prediction and found the RNN model had higher prediction accuracy compared to conventional machine learning models. Bhimireddy et al.[39] compared the performance of conventional BGL prediction models based on LSTM, BiLSTM, and CRNN with sequence-to-sequence (Seq2seq) models and found the Seq2seq models outperform the conventional DNN models. The Seq2seq models generally adopt an encoder-decoder structure where the encoder extracts complex features from input sequences and the decoder recovers future BGL sequences from the obtained deep features. As the future BGLs are predicted simultaneously, the Seq2seq models are more robust compared to conventional BGL forecast models.

Motivated by the representation ability of encoder-decoder structures, Xu et al.[40] proposed a glucose prediction model based on the encoder-decoder structure, where the LSTM was utilized to encode the input information and an attention-based decoder was designed to predict future BGLs. To address the challenge of limited data resources, He et al.[41] developed a personalized autocorrelation encoder that mapped diverse physiological factors into a unified latent space to obtain sparse representations. Considering the potential influence of instability in glucose dynamics, Xue et al.[42] developed a BGL forecasting model based on Informer, where the periodic and trend features were extracted to enhance the prediction capacity of the model. Lee et al.[26] incorporated the Transformer encoder with a multi-task learning framework where temporal correlations in glycemic dynamics were captured by the Transformer encoder and further utilized for BGL prediction and abnormal glycemic events classification. Zhu et al.[27] proposed a light-weight BGL prediction model by integrating LSTM encoder with self-attention mechanism, and the population model enabled a prolonged period on wearable devices. Bian et al.[43] proposed a hybrid BGL prediction model with two encoders, where a stacked LSTM-based encoder was developed to obtain temporal features and a Transformer-based encoder was utilized to enhance long-term dependency. Karagoz et al.[44] conducted a study to investigate the performance of diverse Transformer-based models for BGL forecast against different lengths of history information and found that the best prediction performance was achieved with an input length of one week. Despite the extremely long history of information being utilized for predicting future BGLs, the potential disturbances within the prediction horizon are ignored, which may harm the long-term BGL forecast.

Methods

This work aims to develop a BGL prediction model where the potential disturbances can be encoded into the latent space and transferred through knowledge distillation for improving the model's capacity. Particularly, a Transformer encoder-based teacher model was designed to mine the temporal correlations from the historical observations, together with the future disturbances. The obtained features were transposed to a shared decoder to ensure the latent representations were related to future BGLs. Then, the student model was developed through knowledge distillation to mimic the behaviors of the teacher model based solely on historical observations, therefore, incorporating information of future disturbances into the BGL forecasting model. The overall framework of the proposed BGL prediction model is depicted in Fig. 1.

Feature extraction

The variation of BGL is influenced by many factors, including meal intake, insulin infusion, physical activity, stress, etc. Among those factors, meal intake is the major source of glucose, and BGL usually increases rapidly after carbohydrate consumption. Differently, as the only hormone that promotes the transport, utilization, and storage of plasma glucose, BGL generally decreases after insulin infusion. To establish an accurate BGL forecasting model, the influence of those factors should be taken into account. However, the meal information is usually self-recorded by individuals with diabetes as a sparse impulse signal that happens 3-5 times per day. And

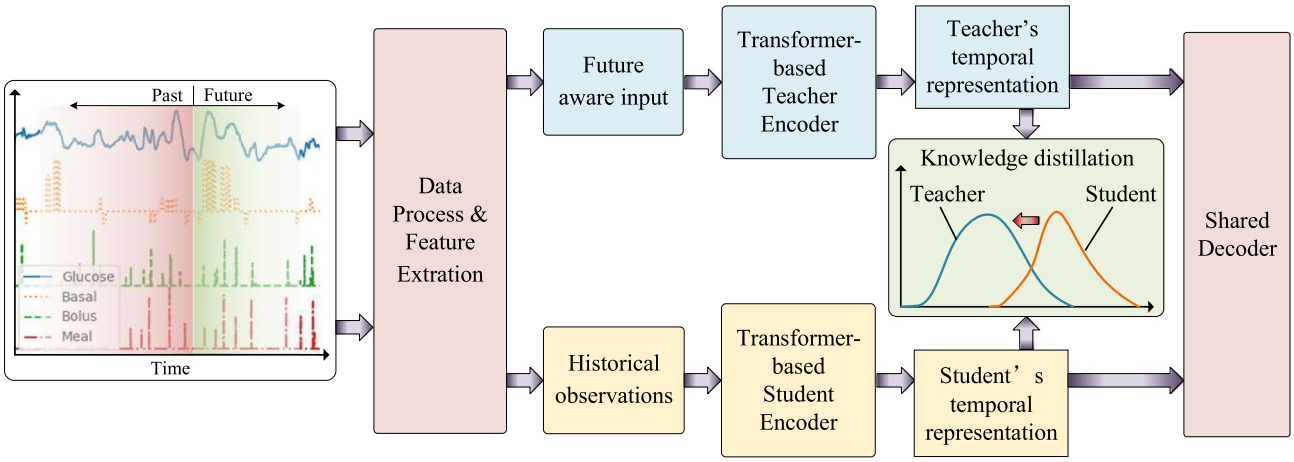


Fig. 1. The framework of the proposed BGL prediction model, consisting of data processing, teacher and student encoders, shared decoders, and knowledge distillation.

the bolus insulin, including the meal and correction bolus, is usually given along with the meal, which is also sparse signals with a similar frequency as meal events. Despite the considerable influence of those sparse factors, it is challenging for the DNN models to learn directly from the sparse signals.

To facilitate the utilization of meal and bolus information, we adopted physiological models to extract advanced features from sparse signals. The insulin on board (IOB), a physiological variable that is defined to describe the amount of active insulin remaining in the body, is utilized to summarize bolus information, as in many hybrid BGL prediction models. The pharmacokinetics of subcutaneously infused insulin are defined as follows[45].

$$\begin{aligned} \dot{S}_1(t) &= u_{bo}(t) - k_{DIA} S_1(t) \\ \dot{S}_2(t) &= k_{DIA}\left(S_1(t) - S_2(t)\right) \\ u_{iob}(t) &= S_1(t) + S_2(t) \end{aligned} \tag{1}$$

where u_{bo} and u_{iob} represent bolus insulin and IOB, respectively, and S_1 and S_2 are the two absorption compartments. k_{DIA} defines the duration of insulin action. A similar physiological model for describing the amount of carbohydrate remaining in the body (COB) is adopted, which is defined as follows[46].

$$\begin{aligned} Ra(t) &= \frac{u_{carb} C_{bio} t \exp -t/t_{max}}{{t_{max}}^2} \\ u_{cob}(t) &= u_{carb} C_{bio} - \int_{t_{meal}}^{t} Ra(t)\, dt \end{aligned} \tag{2}$$

where u_{carb} and u_{cob} are the amount of ingested carbohydrate and remaining active carbohydrate in the body, respectively. The C_{bio} and t_{max} represent the bio-availability and the time of the maximum appearance rate of glucose in the accessible glucose compartment.

Detailed model structures

Teacher encoder

For medium- and long-term glucose prediction, potential disturbances such as carbohydrate intake and insulin administration exert a substantial influence on future blood glucose dynamics. To explicitly leverage the effects of such disturbances during training, a teacher encoder is designed to extract temporal representations from both historical observations x_{past} and future disturbance information x_{future}. The overall structure of the teacher model is illustrated in Fig. 2(a), where Transformer-based encoders are employed to capture long-range temporal dependencies.

The Transformer is a sequence-to-sequence architecture originally developed for natural language processing[47]. It consists of an encoder that learns contextualized representations through self-attention and feed-forward layers, and a decoder that generates outputs by selectively attending to relevant encoded features via cross-attention. Owing to its ability to retain global temporal context and identify informative time steps, the Transformer has demonstrated superior performance over recurrent neural networks for long sequential data. Consistent with prior studies in glucose prediction[26], only the Transformer encoder is utilized in this work. Each encoder comprises an embedding layer, positional encoding, and stacked multi-head attention blocks, followed by normalization and residual connections.

To obtain comprehensive temporal representations from both past observations and future disturbances, two separate Transformer encoders are adopted. Each encoder applies multi-head self-attention to extract high-level temporal features, where the self-attention operation is defined as[47]:

$$\begin{aligned} &\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V, \\ &Q = W_Q X, \\ &K = W_K X, \\ &V = W_V X, \end{aligned} \tag{3}$$

where Q, K, and V denote the query, key, and value matrices obtained from the input X, and W_Q, W_K, W_V are the corresponding weight matrices. d_k is the dimension of the key vectors. To reduce computational complexity, the position-wise feed-forward layers following the attention blocks are replaced with one-dimensional convolutional layers.

Beyond independent encoding, the teacher model further employs a cross-attention mechanism to fuse information from historical observations and future disturbances. Specifically, the encoded historical representations are used as queries, while the encoded future disturbance representations serve as keys and values, allowing the model to selectively attend to future disturbance patterns that are most relevant to the historical glucose context. This cross-attention operation enables the teacher encoder to learn disturbance-aware temporal representations by explicitly modeling the interaction between past physiological states and future exogenous inputs. The fused representations are then aggregated and passed to a shared decoder to generate future blood glucose predictions.

Student encoder

For real-world prediction, the model has access only to historical observations. The student encoder therefore mirrors the teacher encoder but processes only x_{past} (Fig. 2 (b)). It uses the same embedding and sinusoidal positional encoding layers as the teacher to preserve temporal ordering. The embedded inputs pass through identical stacks of multi-head attention and feed-forward blocks with residual connections, producing a historical latent representation.

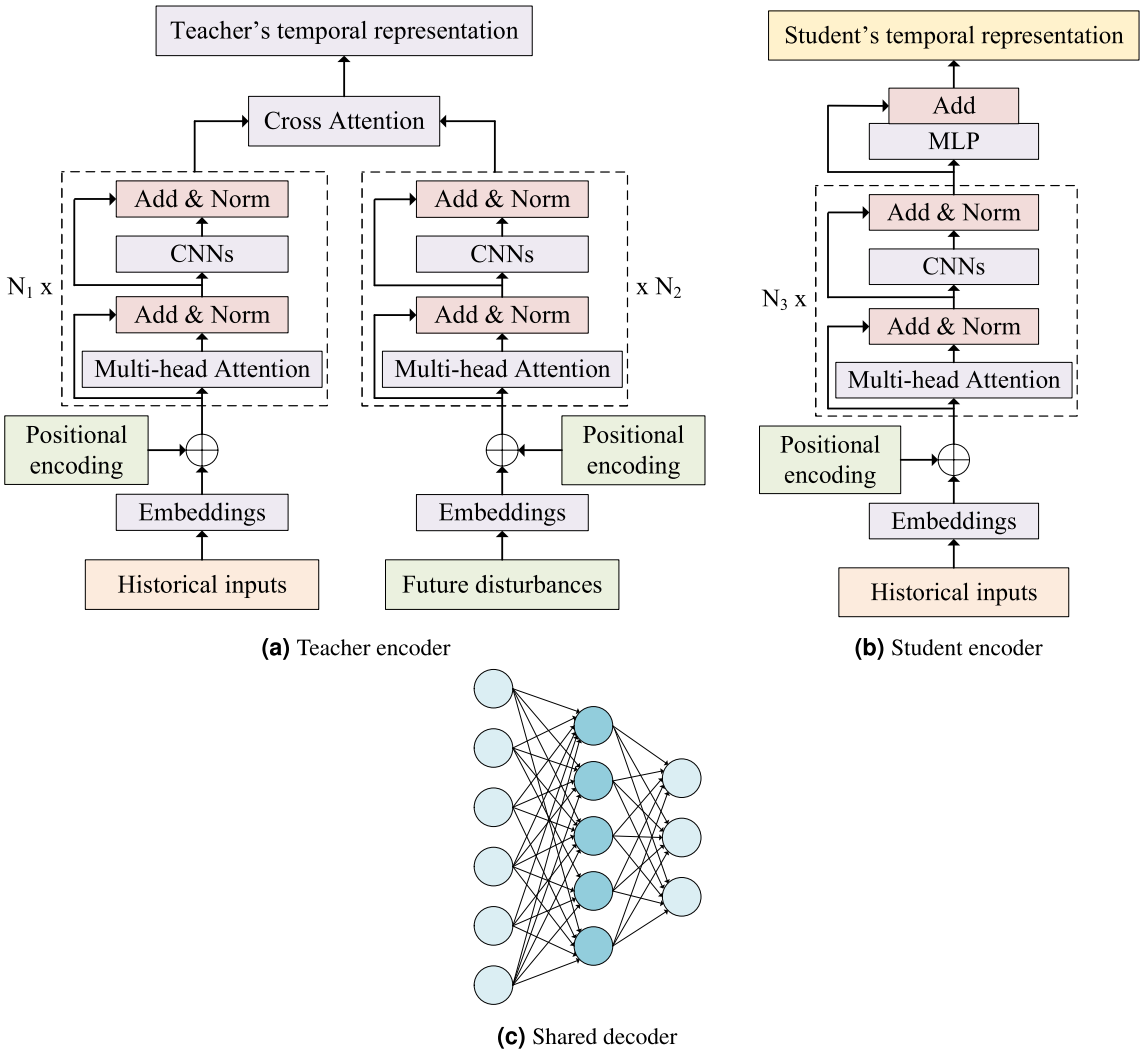


Fig. 2. The detailed structures of the teacher, student encoders and shared decoder.

To ensure the student learns rich temporal features despite missing future information, an additional MLP projection head with residual connections refines its output. This projection head aligns the student's representation space with that of the teacher during knowledge distillation, narrowing the gap between disturbance-aware and historical-only features. In effect, the student encoder learns to approximate the teacher's latent space and capture implicit disturbance effects without direct access to future inputs.

Shared decoder

A single shared decoder (Fig. 2c) maps the latent representations from both the teacher and student encoders to future BGL predictions. The decoder consists of stacked fully connected layers with nonlinear activation (ReLU), dropout for regularization, and a final linear output layer. The shared model ensures that both teacher and student benefit from the same decoding process, improving consistency between training and inference.

Knowledge distillation process

In practical deployment scenarios, future disturbance information such as meal intake and insulin administration is unavailable at inference time, even though it can be observed during offline training. This mismatch between training and deployment conditions limits the performance of conventional blood glucose forecasting models that rely solely on historical observations. To address this issue, we employ a knowledge distillation strategy that enables a deployable student model to internalize information learned under relaxed training assumptions.

The teacher model is trained offline using both historical CGM measurements and future disturbance inputs. By observing future meals and insulin events, the teacher encoder learns disturbance-aware temporal representations that capture how exogenous factors shape subsequent glucose dynamics. These representations encode not only glucose trends but also the anticipated direction, magnitude, and temporal persistence of disturbance-induced effects.

The student model shares the same sequence-to-sequence architecture as the teacher but operates under deployment constraints, using only historical CGM data as input. Since future disturbance information is

unavailable to the student, direct supervision using prediction loss alone is insufficient to transfer the teacher's knowledge. Knowledge distillation is therefore introduced to bridge this information gap.

Rather than distilling output predictions, we focus on transferring latent temporal representations learned by the teacher encoder. Let H^T and H^S denote the hidden representations produced by the teacher and student encoders, respectively. A latent distillation loss is defined to minimize the discrepancy between these representations:

$$L_{KD} = \left\| H^T - H^S \right\|_2^2 \tag{4}$$

This objective encourages the student to approximate the teacher's disturbance-aware internal dynamics using historical information alone. By aligning its latent representations with those of the teacher, the student learns internal structures that reflect the underlying glucose trajectories typically response to following recent historical patterns. These structures implicitly encode expectations about the direction and intensity of upcoming glucose changes, as well as the temporal persistence of their effects, despite the absence of explicit future inputs. As a result, the student is able to approximate disturbance-aware behavior using historical observations alone, thereby mitigating the information loss caused by deployment constraints.

The final training objective of the student model combines the standard prediction loss with the latent distillation loss, forming a joint optimization problem that balances forecasting accuracy and representation alignment, that is

$$\begin{aligned} L &= l_{pred} + \alpha l_{KD} + \beta l_{soft} \\ l_{pred} &= MSE\left(\hat{y}_{Si}, y_i\right) \\ l_{KD} &= MAE\left(x_S, x_T\right) \\ l_{soft} &= MSE\left(\hat{y}_{Si}, \hat{y}_{Ti}\right) \\ x_T &= f_T\left(x_{past}, x_{future}\right) \\ \hat{y}_T &= f_{sdec}\left(x_T\right) \\ x_S &= f_S\left(x_{past}\right) \\ \hat{y}_S &= f_{sdec}\left(x_S\right) \end{aligned} \tag{5}$$

where l_{pred}, l_{KD}, and l_{soft} represent the ground-truth loss, knowledge distillation loss, and "soft label" loss, respectively. And the ground-truth of future BGLs, the BGLs forecast by the student and teacher are defined as y, $\hat{y}_S$, and $\hat{y}_T$, respectively. x_T and x_S are the future disturbance-aware temporal representations of the teacher and incomplete temporal features obtained by the student. The teacher, student, and shared decoder models are defined as f_T, f_S, and f_{sdec}, respectively. Mean squared error was adopted for the prediction-related losses, and mean absolute error was chosen for knowledge distillation, forcing the student to learn same representations as the teacher. The losses are balanced by hyperparameters α and β, which can be determined through cross-validation or grid search. The prediction loss ensures that the student remains directly optimized for multi-step glucose forecasting, while the distillation term guides the learning process toward internal dynamics consistent with those of the disturbance-aware teacher. Through this joint objective, the student model acquires enhanced temporal representations that improve forecasting robustness under realistic inference conditions.

Results

Dataset and preprocess

OhioT1DM dataset

The OhioT1DM dataset[48] contains eight weeks of continuous data from 12 individuals living with T1DM (5 female, 7 male; aged 20 60 years) under free-living conditions. The OhioT1DM dataset was naturally separated into training and testing datasets. The training dataset was utilized for model development and validation, and the testing dataset was adopted for testing the performance of the proposed model. All participants used insulin pump therapy (Medtronic 530G or 630G) and wore Medtronic Enlite CGM sensors, which recorded BGL every 5 minutes. In addition to CGM readings and insulin information, the dataset includes physiological data from fitness bands, such as heart rate, galvanic skin response (GSR), skin temperature, and accelerometry, as well as self-reported events, including meal intake, sleep, work, stress, hypoglycemia, illness, and exercise. The dataset was originally divided into training and testing sets as summarized in Table 1. Basal, bolus, and self-reported meal information were aligned with CGM records at a uniform 5-minute sampling interval.

Because of sensor replacement and communication issues, missing CGM segments occur in both the training and testing sets. In the training data, gaps shorter than 30 minutes were linearly interpolated from surrounding values, and longer gaps were discarded to avoid introducing additional noise. In the test data, the last available CGM value was carried forward until a new measurement became available, ensuring no leakage of future information into the model during evaluation. This pre-processing strategy maintains temporal consistency while minimizing bias from missing data, as done in most related works.

AZT1D dataset

The AZT1D dataset[49] contains data collected from 25 subjects with T1DM who regulated their BGLs with the automated insulin delivery system. Most individuals provide information, including CGM readings (Dexcom G6 Pro) with a sampling time of 5 minutes, insulin administration, and carbohydrate intake, that lasts for over 6-8 weeks, reflecting typical outpatient care practices. In order to evaluate the performance of the proposed method, the first 80% data were utilized for model development and the remaining 20% were keep for testing.

Similar to the OhioT1DM dataset, there are potential missing values in the CGM readings. In this work, constant CGM readings lasting over 30 minutes were treated as missing values. For the training dataset, the gaps

pID	Gender	Age	Training Samples	Testing Samples	Missed CGM during Testing
Released year: 2020					
540	male	20-40	11947	2884	8 stretches, 169 samples
544	male	40-60	10623	2704	6 stretches, 131 samples
552	male	20-40	9080	2352	10 stretches, 433 samples
567	female	20-40	10858	2377	11 stretches, 481 samples
584	male	40-60	12150	2653	15 stretches, 330 samples
596	male	60-80	10877	2731	6 stretches, 258 samples
Released year: 2018					
559	female	40-60	10796	2514	11 stretches, 362 samples
563	male	40-60	12124	2570	3 stretches, 121 samples
570	male	40-60	10982	2745	9 stretches, 134 samples
575	female	40-60	11866	2590	10 stretches, 128 samples
588	female	40-60	12640	2791	2 stretches, 89 samples
591	female	40-60	10847	2760	4 stretches, 86 samples

Table 1. Demographic information of the OhioT1DM dataset.

shorter than 30 minutes were linearly interpolated from surrounding values, and longer gaps were discarded. For testing, the latest available CGM reading was carried forward until a new value became available, preventing information leakage.

Experiment setting

All methods were implemented in PyTorch and executed on a single NVIDIA RTX 5090 D GPU with CUDA version 12.9. The proposed BGL prediction model was compared with several state-of-the-art approaches, including an LSTM model[39], a BiLSTM model[39], a CRNN model[19], a Transformer model[26], and a zero-hold baseline where the most recent CGM value was used as the forecast. In addition, we examined the performance of (i) the traditional Seq2One regression modeling strategy, which predicts a single future target value from historical observations, and (ii) the Seq2seq modeling strategy, which forecasts the entire sequence of future targets simultaneously.

For both the teacher and student encoders, input sequences were embedded into a 64-dimensional latent space. The number of multi-head attention units was set to 4. Each encoder block contained two 1D CNN layers, where features were firstly projected to 128 dimensions and then mapped back to 64. The number of encoder blocks was fixed at $N_1 = N_2 = N_3 = 3$, determined through cross-validation on the training set. The hyperparameters α and β are jointly searched within [0.8, 0.5, 0.2, 0.1] and [0.5, 0.2, 0.1, 0.05, 0.01]. The shared decoder followed the same structure as in the CRNN model[19].

The glucose values reported by CGM, estimated IOB and COB within the 3-hour historical window were used to predict future BGL values up to 2 hours as a simultaneous sequence. The input and output of the models are normalized through min-max scaling. Models were trained with a batch size of 512, a learning rate of 0.0001 using the Adam optimizer, and an early stopping criterion of no improvement in validation loss for 20 epochs or reaching the maximum training steps 200. All models were initially trained on the population data and subsequently fine-tuned on individual data to enable personalized BGL prediction.

Experimental results

The prediction accuracy of the BGL prediction models was evaluated by root mean squared error (RMSE), mean absolute error (MAE) and the Clarke error grid analysis, which are common metrics used in the field. The RMSE and MAE are defined as follows.

$$\begin{aligned} RMSE &= \sqrt{\frac{\sum_{i=1}^{N}(\hat{y}_i - y_i)^2}{N}} \\ MAE &= \frac{\sum_{i=1}^{N}|\hat{y}_i - y_i|}{N} \end{aligned} \tag{6}$$

where $\hat{y}$ and y are the forecast and measured BGLs and N is the total number of samples. The Clarke error grid analysis divides the distribution of measured glucose levels and estimated values into five zones based on clinical relevance, from clinically accurate to significant error. where $\hat{y}$ and y are the forecast and measured BGLs and N is the total number of samples. The Clarke error grid analysis divides the distribution of measured glucose levels and estimated values into five zones based on clinical relevance, from clinically accurate to significant error.

Results for OhioT1DM dataset

The averaged value and standard deviation of RMSE (mg/dL) and MAE (mg/dL) across all 12 subjects for different BGL prediction models are summarized in Table 2 and 3. For all the investigated prediction horizons (PHs), the BGL forecast model based on DNN methods outperforms the zero-hold method in terms of RMSE and MAE. Among the Seq2One models, the BiLSTM model outperforms the LSTM model for short-term

Models		PH=30 min		PH=60 min	
		RMSE	MAE	RMSE	MAE
Seq2One	Zero-hold	23.32±2.78	17.01±1.98	38.41±4.47	28.68±3.30
	LSTM	18.63±2.20	13.14±1.47	32.73±3.42	24.26±2.67
	BiLSTM	18.52±2.16	13.09±1.46	32.02±3.58	23.65±2.91
	CRNN	19.08±1.91	13.53±1.38	31.98±3.88	23.65±3.19
	Transformer	18.34±2.16	12.98±1.51	31.15±3.35	22.88±2.65
Seq2seq	LSTM	18.77±2.22	13.20±1.43	31.85±3.62	23.27±2.76
	BiLSTM	18.81±2.51	13.26±1.58	31.56±3.64	23.21±2.85
	CRNN	18.57±2.38	13.09±1.54	30.99±3.39	22.87±2.69
	Transformer	18.48±2.26	13.10±1.45	30.98±3.39	22.85±2.56
Proposed	Teacher	$17.83 \pm 2.27^{*}$	$12.73 \pm 1.43^{*}$	$28.81 \pm 2.97^{*}$	$21.50 \pm 2.17^{*}$
	Student	**18.24±2.33**	**12.91±1.47**	**30.47±3.39**	**22.54±2.82**

Table 2. The RMSE and MAE (Mean±STD) of glucose prediction models based on neural networks for short-term prediction with OhioT1DM dataset. *p<0.01, **p<0.05

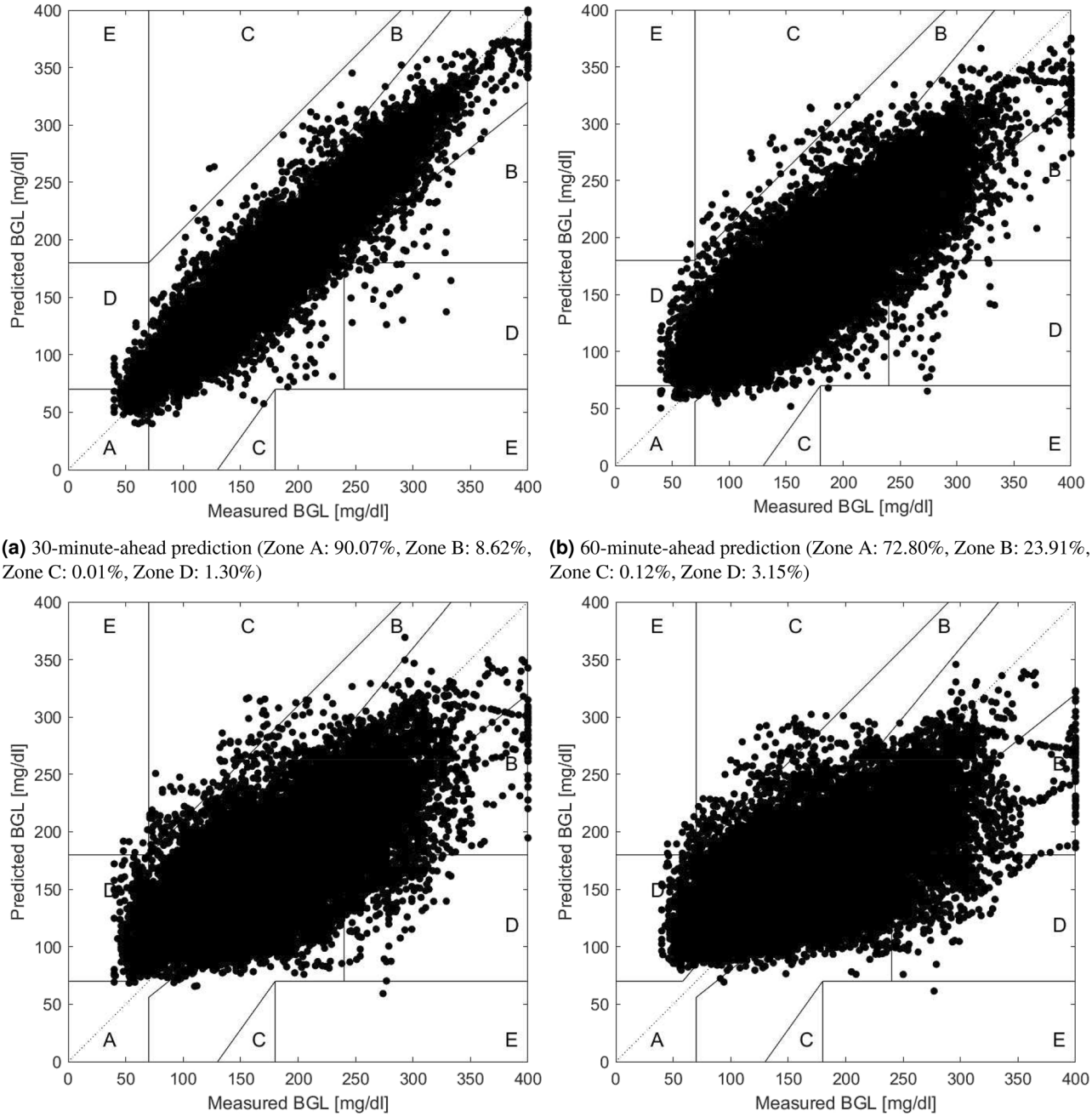
Models		PH=90 min		PH=120 min	
		RMSE	MAE	RMSE	MAE
Seq2One	Zero-hold	49.21±5.53	37.28±4.09	57.11±6.13	43.77±4.52
	LSTM	41.02±3.76	31.24±3.18	46.75±4.47	36.33±3.61
	BiLSTM	41.29±3.85	31.44±3.07	46.98±4.59	36.70±3.66
	CRNN	41.00±4.25	31.13±3.46	46.74±4.28	36.29±3.43
	Transformer	40.71±4.29	30.98±3.38	46.30±4.67	36.09±3.72
Seq2seq	LSTM	41.28±4.53	30.97±3.49	47.61±4.83	36.46±3.81
	BiLSTM	40.99±4.55	30.99±3.57	47.28±5.34	36.51±4.17
	CRNN	40.00±4.05	30.43±3.36	46.38±4.55	36.02±3.70
	Transformer	40.03±4.17	30.30±3.22	46.26±4.77	35.67±3.64
Proposed	Teacher	$36.38 \pm 3.92^{*}$	$27.68 \pm 3.01^{*}$	$41.93 \pm 4.55^{*}$	$32.42 \pm 3.63^{*}$
	Student	$\mathbf{39.51 \pm 3.97}^{**}$	$\mathbf{29.95 \pm 3.42}^{**}$	$\mathbf{45.60 \pm 4.53}^{**}$	$\mathbf{35.28 \pm 3.71}^{**}$

Table 3. The RMSE and MAE (Mean±STD) of glucose prediction models based on neural networks for long-term prediction with OhioT1DM dataset. *p<0.01, **p<0.05

glucose prediction, and the consideration of local patterns and temporal features further improves the prediction accuracy, except for 30-minute-ahead forecast. Different from the conventional DNN models, the Transformer model retains information of the entire input sequence, thus achieving the lowest prediction error. Compared to the conventional Seq2One glucose forecast models, the Seq2seq models generally have smaller RMSEs and MAEs, indicating that those models outperform the classical ones in BGL prediction. For the conventional RNN models, the prediction errors are slightly smaller for medium-term forecasting (i.e., PH=90 and 120 min), while for PH=30 and 120 min, the prediction errors are greater compared to the Seq2One strategy. For the models with a more complex structure, the Seq2seq model has more accurate predictions, except for the case of PH=30 min, where the prediction error increases slightly.

When the future disturbance information is announced to the BGL prediction model, a considerable improvement in the model forecast performance can be observed, where the RMSEs decrease from 18.48, 30.98, 40.03, and 46.26 to 17.83, 28.81, 36.38, and 41.93 *mg*/*dL* for PH=30, 60, 90, and 120 minutes, respectively. Similarly, the MAEs also have a significant improvement. Compared to the teacher model, as the student model cannot observe disturbances directly, the forecast ability of the student model is weaker. Despite the weakness compared to the teacher, the student model still has lower prediction errors in contrast to all investigated Seq2seq and Seq2One models that ignore the influence of future disturbance on the variation of glucose dynamics for both short- and long-term predictions.

The BGL values reported by the CGM sensors and forecast by the proposed model for different prediction horizons are compared in Fig. 3. Zone A and Zone B represent the regions where the predicted values can be utilized for diabetes management without causing safety issues. Zones C and D are the over- or under-estimate regions that may lead to overdose or underdose of insulin therapy. Zone E indicates a clinical error and cannot be used for clinical decision-making. For all the investigated prediction horizons, over 90% of estimated results of the proposed method are in the clinically accurate and acceptable regions, indicating the model can meet the requirement for diabetes management.



(a) 30-minute-ahead prediction (Zone A: 90.07%, Zone B: 8.62%, Zone C: 0.01%, Zone D: 1.30%)

(b) 60-minute-ahead prediction (Zone A: 72.80%, Zone B: 23.91%, Zone C: 0.12%, Zone D: 3.15%)

(c) 90-minute-ahead prediction (Zone A: 60.70%, Zone B: 34.46%, Zone C: 0.32%, Zone D: 4.50%)

(d) 120-minute-ahead prediction (Zone A: 52.87%, Zone B: 40.86%, Zone C: 0.47%, Zone D: 5.74%)

Fig. 3. Clarke error grid analysis of the proposed method with OhioT1DM dataset.

Results for AZT1D dataset

The prediction error metrics of the proposed method with the AZT1D dataset are summarized in Table 4 and 5. Compared to the baseline Seq2seq models, the incorporation of future disturbances improves RMSE from 19.49 *mg/dL* to 17.36 *mg/dL* for 30-minute-ahead prediction, 29.96 *mg/dL* to 24.51 *mg/dL* for 60-minute-ahead prediction. As the increase of PH, the RMSEs are 28.40 and 31.91 *mg/dL* for 90- and 120-minute-ahead prediction, significantly lower in contrast to the baseline models. Similar for the MAE, the improvements are 1.52, 4.27, 6.33, and 6.58 *mg/dL* for 30- to 120-minute-ahead prediction, respectively.

The Clarke error grid analysis of the proposed model with AZT1D dataset is summarized in Fig. 6. For all the investigated PHs, there are over 90% of forecasted BGL values are distributed in the clinical acceptable zones. As the increase of PH, the percentage of forecasted BGLs in the clinical dangerous zones increases slightly, from 1.41% for 30-minute-ahead prediction to 3.53% for 120-minute-ahead prediction. Overall, the results demonstrate that the proposed model enables prediction of future BGLs with high clinical acceptable rate.

Models		PH=30 min		PH=60 min	
		RMSE	MAE	RMSE	MAE
Seq2seq	LSTM	19.38±3.35	13.98±2.64	29.85±4.86	22.51±3.91
	BiLSTM	19.30±3.38	13.88±2.66	29.71±4.90	22.30±3.94
	CRNN	19.40±3.20	14.01±2.55	29.88±4.75	22.56±3.84
	Transformer	19.49±3.40	14.05±2.68	29.96±4.95	22.52±3.99
Proposed	Teacher	$17.36 \pm 2.84^{*}$	$12.53 \pm 2.19^{*}$	$24.51 \pm 3.73^{*}$	$18.25 \pm 2.94^{*}$
	Student	**19.38±3.28**	**13.97±2.60**	**29.77±4.72**	**22.43±3.85**

Table 4. The RMSE and MAE (Mean±STD) of glucose prediction models based on neural networks for short-term prediction with AZT1DM dataset. *p<0.01, **p<0.05

Models		PH=90 min		PH=120 min	
		RMSE	MAE	RMSE	MAE
Seq2seq	LSTM	36.07±6.34	27.79±5.06	39.80±7.54	30.93±5.99
	BiLSTM	36.06±6.40	27.72±5.09	39.87±7.75	31.10±6.10
	CRNN	36.10±6.25	27.84±4.98	39.76±7.43	31.09±5.97
	Transformer	36.06±6.33	27.75±5.08	39.68±7.53	30.85±6.06
Proposed	Teacher	$28.40 \pm 4.82^{*}$	$21.42 \pm 3.69^{*}$	$31.91 \pm 5.87^{*}$	$24.27 \pm 4.38^{*}$
	Student	$\mathbf{35.90 \pm 6.07}^{**}$	$\mathbf{27.54 \pm 4.95}^{**}$	$\mathbf{39.47 \pm 7.18}^{**}$	$\mathbf{30.66 \pm 5.86}^{**}$

Table 5. The RMSE and MAE (Mean±STD) of glucose prediction models based on neural networks for long-term prediction with AZT1DM dataset. *p<0.01, **p<0.05

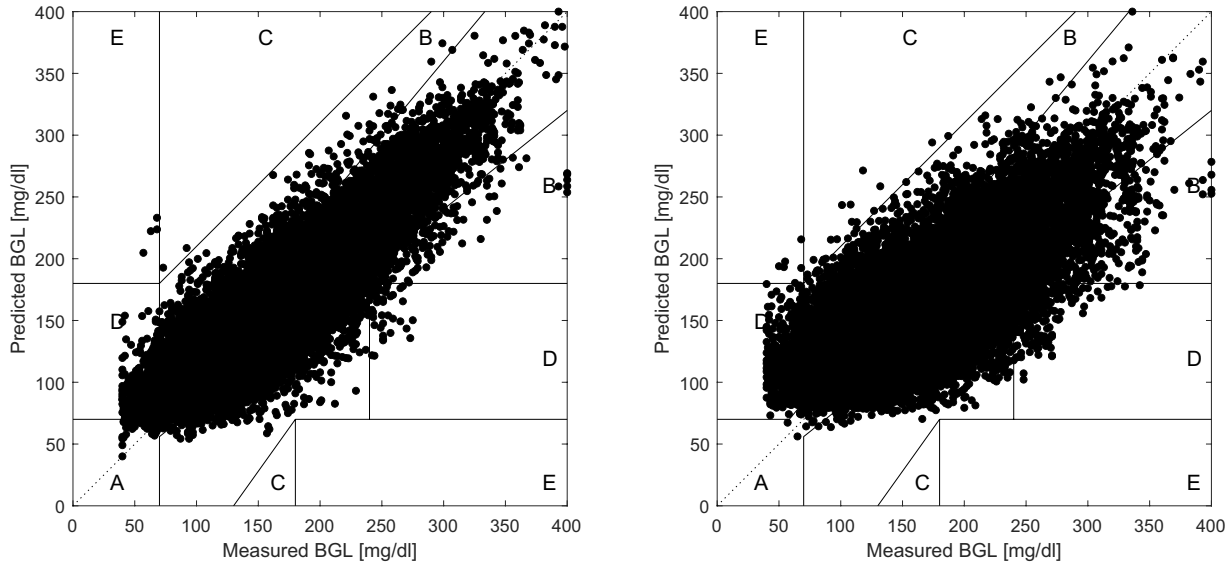
Discussion

Seq2One and Seq2Seq are the two major modeling strategies in blood glucose forecasting, where target values are predicted separately or simultaneously. Among the Seq2One models investigated in this study, the Transformer-based model achieved the lowest prediction error (RMSE and MAE) for short- to long-term predictions, while the CRNN performed worst in short-term prediction. This may be because local features were pooled in CRNN, losing patterns critical for short-term forecasts. Similarly, bidirectional information in the BiLSTM did not improve medium- to long-term prediction, as reflected in its larger RMSEs and MAEs. By contrast, Seq2Seq models generally improved prediction accuracy, as simultaneous prediction of future BGLs forces the models to generalize over the entire sequence. However, we also observed some decline in short-term forecast ability with RNN-based Seq2Seq models despite their overall advantage in long-term predictions.

From a clinical perspective, the infusion of subcutaneous insulin and the consumption of meals are among the strongest drivers of glucose excursions. Forecasting BGL without explicitly modeling these disturbances raises the risk of under- or over-estimating future glucose levels, which can lead to missed opportunities for corrective action or unnecessary interventions. By incorporating future disturbance information into the modeling process, our Transformer-based Seq2Seq model achieved significant improvements in short- to long-term predictions (RMSE improved by 0.65-4.33 mg/dL and MAE improved by 0.37-3.25 mg/dL with OhioT1DM dataset, RMSE improved by 2.11-7.64 mg/dL and MAE improved by 1.48-6.58 mg/dL with AZT1D dataset). The results indicate that disturbance-aware modeling particularly benefits medium- to long-term forecasts, even though future insulin and meal information is not inherently available in the historical data. Clinically, more accurate medium- to long-term forecasts can enable better timing of insulin boluses, meal planning, and proactive prevention of hypo- or hyperglycemia.

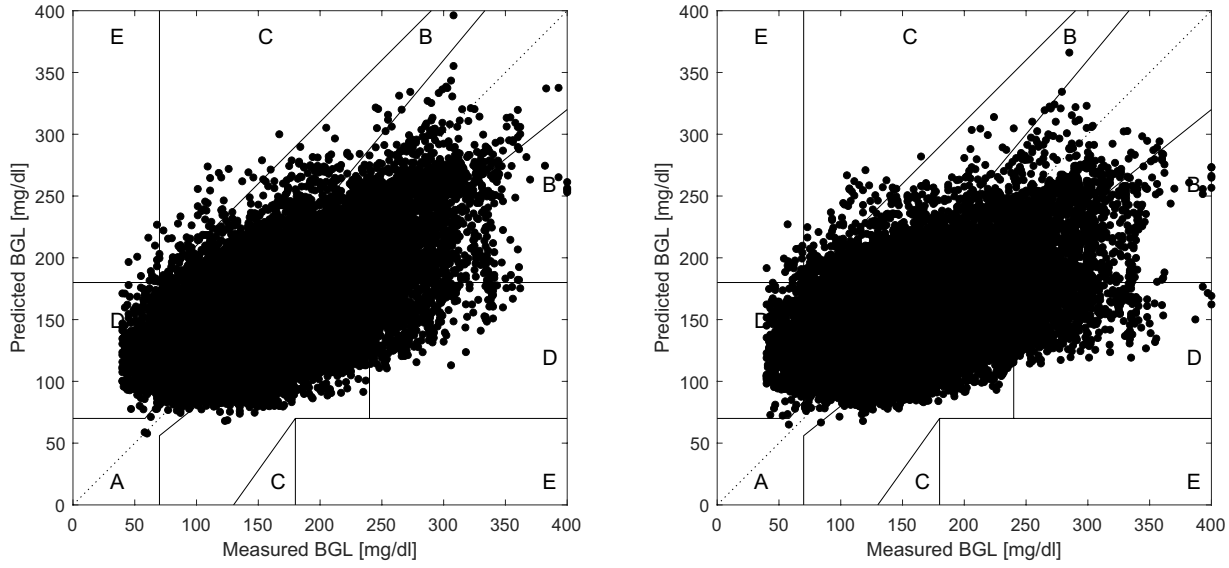
When the knowledge obtained by the teacher encoder was transferred to the student model, the student, who lacks direct future disturbance inputs, was able to approximate disturbance-aware patterns and improve BGL forecasts. This demonstrates the practical deployability of our framework, that is, the student model can be implemented in real-world systems (e.g., decision-support or closed-loop insulin delivery systems) where only historical data is available, yet still benefits from knowledge distilled from richer datasets during training. As expected, for short-term prediction, the variables that may influence the changes of future BGLs have already been included in the model input, the improvement of model performance is not significant. However, as the prediction horizon becomes larger, a considerable difference can be observed in the prediction metrics, demonstrating the importance of incorporating future disturbance for long-term glucose forecasting.

Figures 4 and 5 compared the predicted and measured BGL trajectories for 30- and 60-minute-ahead forecasts in subjects with the least and most missing gaps. For 30-minute-ahead prediction, the model closely tracked CGM measurements when sensor data was available. However, larger prediction errors emerged during missing CGM segments, which is expected given that zero-hold extrapolation is a weak imputation strategy. We also observed that prediction errors were larger when CGM trajectories saturated, and that sensor noise became more influential at longer horizons, even though predicted values continued to follow the measurements overall.



(a) 30-minute-ahead prediction (Zone A: 86.86%, Zone B: 11.72%, Zone C: 0, Zone D: 1.41%)

(b) 60-minute-ahead prediction (Zone A: 71.02%, Zone B: 26.84%, Zone C: 0.03%, Zone D: 2.10%)



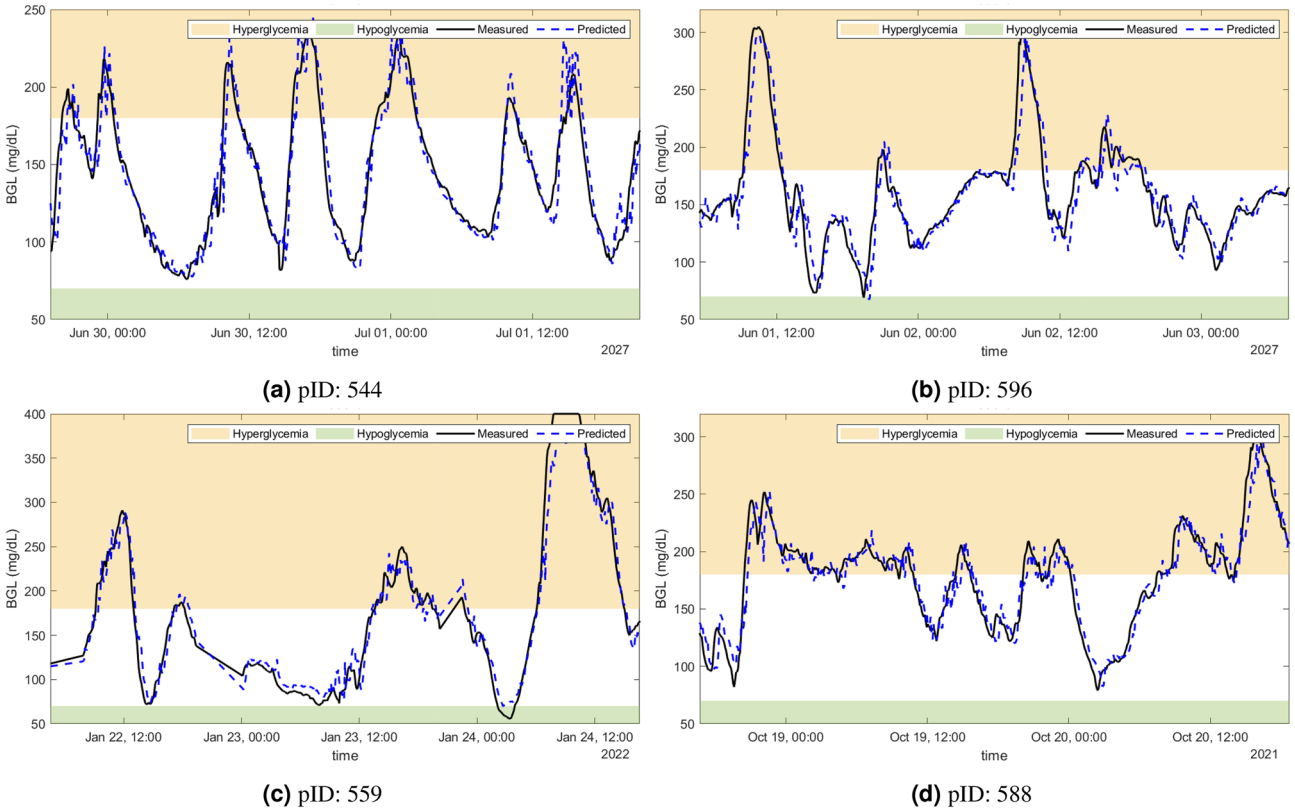
(c) 90-minute-ahead prediction (Zone A: 61.94%, Zone B: 35.17%, Zone C: 0.12%, Zone D: 2.74%)

(d) 120-minute-ahead prediction (Zone A: 56.57%, Zone B: 39.86%, Zone C: 0.15%, Zone D: 3.38%)

Fig. 6. Clarke error grid analysis of the proposed method with AZT1D dataset.

The OhioT1DM dataset, released for the 2018 and 2020 BGL prediction challenges, has fostered significant progress in glucose forecasting. Table 6 summarizes published results ranging from conventional DNN models to advanced deep learning techniques. For short-term prediction (PH = 30 minutes), reported RMSEs and MAEs range from 18.32-20.57 mg/dL and 12.85-25.09 mg/dL, respectively. Incorporating reinforcement learning with a DNN model achieved the lowest forecast error at an RMSE of 18.32 mg/dL and MAE of 12.93 mg/dL across both datasets. In comparison, our proposed model achieved an RMSE of 18.24 mg/dL and MAE of 12.91 mg/dL, slightly outperforming previously published results. For 60-minute-ahead prediction horizons, our model's improvement was even more pronounced. This consistent performance at longer horizons benefits from the knowledge distilled from the teacher and is clinically meaningful because it provides patients and clinicians with actionable insight over extended time frames, enabling proactive interventions.

Accurate medium- and long-term blood glucose forecasting is critical for guiding insulin dosing, meal planning, and hypoglycemia prevention in individuals with diabetes. By transferring future-aware representations



(a) pID: 544 **(b)** pID: 596

(c) pID: 559 **(d)** pID: 588

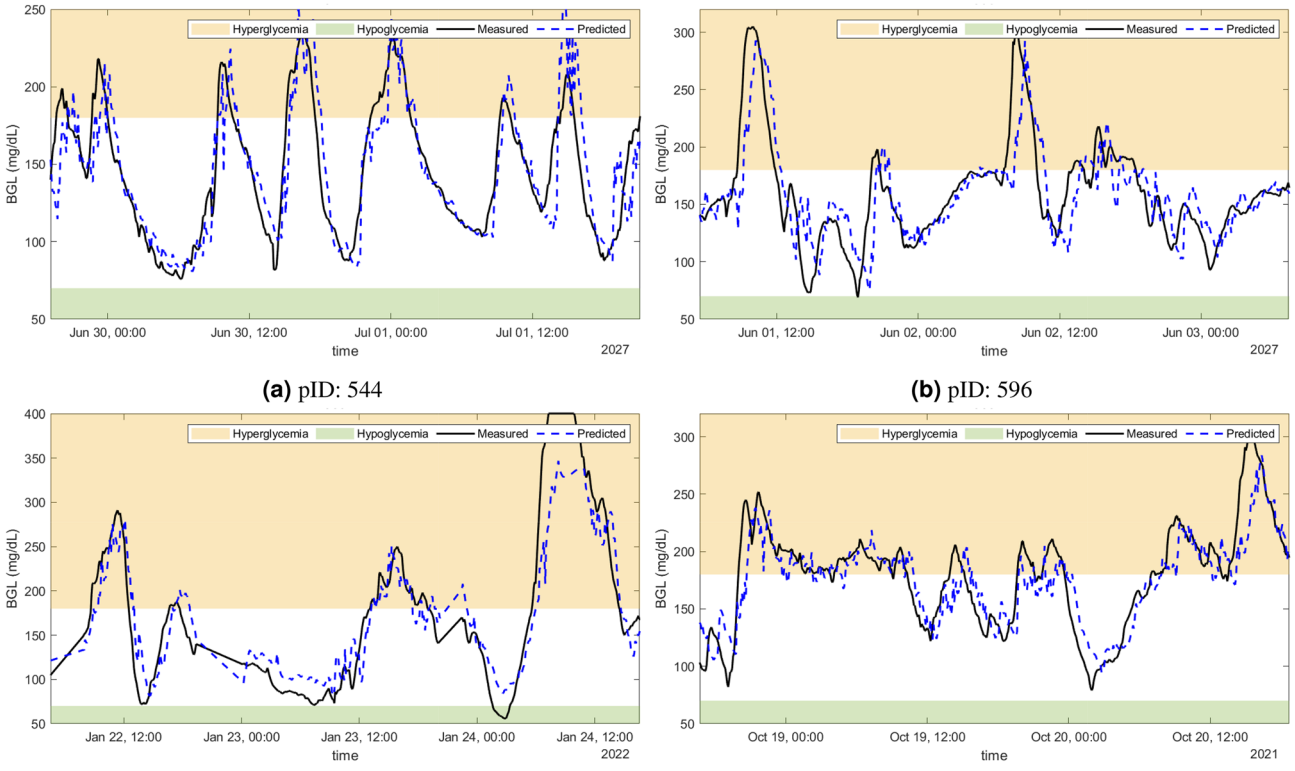
Fig. 4. Illustration of forecasting trajectory of the proposed method for 30-minute-ahead prediction with OhioT1DM dataset.

from a teacher model to a student model usable with only historical inputs, the proposed framework brings advanced predictive capability into settings where real-time future disturbance data are unavailable. This approach could enhance clinical decision-support systems and inform closed-loop insulin delivery, ultimately improving safety, reducing adverse events, and supporting more proactive diabetes self-management.

Despite the advantages of our proposed knowledge-distillation-based model, several limitations remain. The proposed model assumes that the individuals follow a regular daily routine. Even though it is the underlying assumption for predictive tasks, the behavior of the individual may change considerably. To mitigate the influence of abnormal patterns, future work will focus on incorporating announced disturbances into the model. Besides, to ensure fair comparisons, the model size was kept comparable to baselines, which may limit the full potential of the teacher-student framework. Future work will evaluate larger architectures and data augmentation techniques to enhance performance.

Conclusion

In this work, we developed a Transformer-based deep learning framework for short- to long-term blood glucose prediction that incorporates knowledge distillation to exploit future disturbance information during training. The teacher model learns disturbance-aware temporal representations by modeling the influence of anticipated events, such as meal intake and insulin delivery, on glucose dynamics, while the student model is trained to approximate these representations using historical data only. This design enables the student to capture future-relevant features without requiring access to future information at inference time. Experiments conducted on the OhioT1DM and AZT1D datasets demonstrate that the proposed framework yields consistent improvements over baseline deep learning models across multiple prediction horizons. The results suggest that knowledge distillation can serve as a practical mechanism for transferring future-aware representations to deployable models constrained to historical inputs. From a clinical perspective, the observed improvements are more pronounced for medium- to long-term forecasts, which may support proactive decision-making such as insulin adjustment and hypoglycemia risk management. Future work will investigate the incorporation of additional announced information and the evaluation of the proposed framework in real-time and closed-loop insulin delivery settings.



(a) pID: 544 **(b)** pID: 596

(c) pID: 559 **(d)** pID: 588

Fig. 5. Illustration of forecasting trajectory of the proposed method for 60-minute-ahead prediction with OhioT1DM dataset.

Method	published year	dataset	RMSE	MAE
PH = 30 min				
GLuNet[33]	2019	2018	19.28±2.76	–
TCN[38]	2020	2018	19.97±2.83	–
PBGTAM[20]	2023	2018	20.57	25.09
RNN+KD[50]	2020	2020	19.21	13.07
RCN+ARMA[51]	2020	2020	20.03	15.17
CRNN+Multitask learning[52]	2022	2018+2020	18.8±2.8	13.2±2.2
E3NN[18]	2022	2018+2020	18.92±2.12	13.46±1.49
FCNN[53]	2022	2018+2020	18.64±2.60	13.25±1.67
GluGAN[54]	2023	2018+2020	19.8±2.35	14.1±1.55
EPS-TFT[27]	2024	2018+2020	19.1±2.5	13.1±1.6
Deep RL[55]	2024	2018+2020	18.32	12.93
GRU+DA[56]	2024	2018+2020	20.48	14.59
iTDJL[57]	2025	2018+2020	18.58	12.85
GATv2+GRU[58]	2025	2018+2020	18.97±2.43	13.34±1.68
ours	–	2018	**17.83±2.08**	**12.93±1.60**
	–	2020	**18.65±2.50**	**12.90±1.33**
	–	2018+2020	**18.24±2.33**	**12.91±1.47**
PH = 60 min				
GLuNet[33]	2019	2018	31.83±3.49	–
RNN+KD[50]	2020	2020	31.77	23.09
RCN+ARMA[51]	2020	2020	34.89	26.41
CRNN+Multitask learning[52]	2022	2018+2020	32.3±3.9	23.6±3.0
E3NN[18]	2022	2018+2020	32.54±3.61	24.05±2.94
FCNN[53]	2022	2018+2020	31.07±3.62	22.86±2.89
GluGAN[54]	2023	2018+2020	33.43±3.56	24.80±2.72
GRU+DA[56]	2024	2018+2020	34.16	25.54
EPS-TFT[27]	2024	2019+2020	32.3±3.8	23.2±2.8
iTDJL[57]	2025	2018+2020	31.12	22.50
ours	–	2018	**30.76±3.96**	**22.98±3.38**
	–	2020	**30.17±2.67**	**22.10±2.02**
	–	2018+2020	**30.47±3.39**	**22.54±2.82**
PH = 120 min				
CRNN+Multitask learning[52]	2022	2018+2020	47.2±4.6	36.5±3.8
BiLSTM[59]	2021	2020	55.70±2.98	43.76±1.81
Dual LSTM[59]	2021	2020	63.79±3.53	48.63±1.73
Dilated RNN[59]	2021	2020	56.41±2.31	45.63±1.66
Physiological DNN[59]	2021	2020	54.70±2.27	45.05±1.79
ours	–	2018	**46.74±5.27**	**36.60±4.04**
	–	2020	**44.47±3.28**	**33.95±2.77**
	–	2018+2020	**45.60±4.53**	**35.28±3.71**

Table 6. Summary of BGL prediction accuracy for OhioT1DM dataset.

Data availability

The publicly available OhioT1DM dataset used in this research is accessible, followed by requesting a data use agreement at https://ohio.qualtrics.com/jfe/form/SV_02QtWEVm7ARIKIl. The AZT1D dataset used in this res earch is publicly available at https://data.mendeley.com/datasets/gk9m674wcx/1.

Code availability

The code used for reproducing the results of this manuscript is available at https://github.com/Xiaoyu-SUN-NEU/Distrubance-Aware-BGLP.git.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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